Lab 0 ticket

Computers Speak Vectors — Friday, August 28

Lab 0 — Computers Speak Vectors · Fri Aug 28 · Laptop required · Handwrite on this hard copy · Get initials after check-in

Name: _____ **Language (circle):** Octave / Python

New syntax (fill as you go)

I needed to…	Command / pattern (your language)
Create vector `x`	
Length `n`	
Print 3rd Boyd entry	
Change an entry	
Scale / add vectors	
Plot	

Data: $x = (52, 51, 53, 55, 58, 60)$ °F, hours 7 a.m.–noon. Boyd $x_1 \ldots x_6$. Octave: 1-based. Python: 0-based. Do **C0.1–C0.6 in order.**

C0.1 — Enter and index. Create `x`. Print n, first, last, Boyd x_3.

Expected: $n = 6$, first 52, last 60, third 53.

Yours: n =_____ first=_____ last=_____ third=_____

Command that prints the third entry: _____

x_3 means (quantity + units): _____

C0.2 — Modify. Set 9 a.m. to 54, noon to 61. Print new `x`.

Expected: $(52, 51, 54, 55, 58, 61)$. **Yours:** _____

C0.3 — Scale. `doubled = 2*x` (use corrected `x`). Print `doubled`.

Expected: $(104, 102, 108, 110, 116, 122)$.

Assignment line: _____

Meaning of $\times 2$ here: _____

C0.4 — Add a vector. `ones6` = six ones; `warmer = x + 5*ones6`. Print `warmer`.

Expected: $(57, 56, 59, 60, 63, 66)$. **Yours:** _____

Why not use math-course "`x+5`" as vector addition? _____

C0.5 — Plot corrected `x`, `2*x`, and `-x` vs $i = 1..6$ (legend).

What did $\times(-1)$ do in the picture? _____

C0.6 — Predict, then check. Before running: Octave `v=[10,20,30]; v(2)` or Python `v=np.array([10,20,30]); print(v[1])`.

Expected: 20. Prediction: _____ Actual: _____ Boyd index: v___

Stuck on (optional): C0.___ _____